

Nikhita Bonu

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PROFESSIONAL SUMMARY

AI/ML Engineer & Full-Stack Developer with 7+ years of experience delivering scalable enterprise applications and intelligent systems, specializing in conversational AI, LLMs, and real-time event-driven platforms. Proficient in Java, Python, React, and microservices architecture, driving 50% performance improvements in distributed systems. Skilled in ML, Computer Vision, and NLP using PyTorch and TensorFlow to build advanced healthcare imaging and LLM models. Experienced in CI/CD, database optimization, and cloud deployment, delivering end-to-end projects and driven cross-functional collaboration.

EDUCATION

Texas Tech University, Texas, USA

Jan 2025 – May 2026(Expected)

Master's in Computer Science

GPA: 3.66/4.00

Coursework: Machine Learning, Neural Networks, Applied Data Science, Generative AI, Analysis of Algorithms, DBMS.

PROFESSIONAL EXPERIENCE

Erevna Games – Lubbock, USA

Feb 2026 – Present

Product Development Intern

- Conducted end-to-end testing of a **generative AI gaming platform**, validating gameplay mechanics, rules, and AI features for reliability.
- Designed and optimized **Microsoft Copilot** integrations to automate workflows, enhance game logic, and streamline user interactions.
- Developed **user-targeted notification systems** using AI insights and **demographic data**, increasing user engagement by **30%**.

Texas Tech University – Remote, USA

Mar 2025 – Feb 2026

AI/ML Engineer

- Conducted AI-driven analysis of LFA test kits, **automating test/control lines detection from video data** and reducing manual review effort by 50% using advanced computer vision techniques.
- Applied SAM2(Segment Anything-2)** for precise **strip-region segmentation**, improving **feature extraction accuracy by 30%**, and validated results against human interpretation using key diagnostic performance metrics.
- Designed and deployed scalable **n8n automation workflows** and REST API integrations, integrating OpenAI, Google Sheets, CRMs, and databases, **cutting turnaround time by 40%** and improving operational efficiency by 35%.
- Engineered **LLM & RAG-based AI agents** with advanced **prompt engineering**, enabling autonomous task execution and dynamic decision-making, increasing automation coverage by 60% and streamlining reporting pipelines.

IVY Comptech (Entain) - Hyderabad, India

Jan 2021 – Jan 2025

Software Engineer

- Proficient in designing and developing **RESTful microservices using Java** (Spring Boot) and integrating AI/ML inference APIs to deliver intelligent, scalable backend functionalities in distributed systems.
- Optimized SQL, MongoDB, and MySQL queries, **reducing query execution time by up to 90%** and improving backend response latency, while building scalable data-processing scripts to support end-to-end machine-learning workflows.
- Developed responsive frontend components with React.js, **integrating backend services** and implementing **AI-enhanced UI features** such as **recommendations and dynamic data-driven interactions** enhancing the user experience.
- Established end-to-end **CI/CD pipelines with Jenkins, GitHub Actions, Docker, and Kubernetes** automating builds, Junit testing, deployments, **reducing deployment cycles by 30%** and improving release reliability through active code in Agile/Scrum environments.

Aspirevision Tech Education - Hyderabad, India

Jan 2019 – Dec 2020

Machine Learning Project Intern

- Hands-on experience with **machine learning libraries** such as Scikit-Learn, TensorFlow, and PyTorch for **data preprocessing, EDA**.
- Applied supervised, semi supervised and unsupervised algorithms including linear regression, decision trees, k-means clustering for predictive modeling.
- Explored **AWS cloud platforms** for scalable **machine learning workflows**, including model deployment, and performance optimization.

PROJECTS

Local Network-Security Tutor & Quiz Bot

Sep 2025 – Dec 2025

- Designed a secure, privacy-preserving **RAG pipeline by building a local LLM system** with document chunking, Sentence-Transformer embeddings, and FAISS/Qdrant **vector search to support high-accuracy** offline Q&A.
- Orchestrated **multi-agent workflows** including a Q&A Tutor Agent (retrieval + citation generation) and a Quiz Agent (auto-generation + grading of MCQ/T/F/open-ended questions), demonstrating strong skills in **agentic AI systems and prompt orchestration**.
- Implemented an end-to-end **LLM architecture using open-source models**, embedding models, **retrieval modules**, and a user-facing interface showcasing expertise in **LLM integration, system design**, and production-ready AI applications.
- Technologies:** Python, GPT4All/Llama, Transformers, FAISS/Qdrant, Streamlit, RAG, Vector Databases.

Named Entity Recognition with BERT in PyTorch

Mar 2025 – May 2025

- Conducted comprehensive **data preprocessing and label encoding** to prepare raw text for training, ensuring high-quality inputs for the NER pipeline.
- Implemented **custom tokenization** and handling of subword handling using **Hugging Face's BERT tokenizer** to preserve entity boundary accuracy.
- Developed and fine-tuned a BERT-based NER system within a PyTorch framework, achieving a **state-of-the-art F1 score of 0.9108**.
- Performed **hyperparameter tuning and qualitative error analysis** to further optimize model performance and interpretability.
- Technologies:** Python, PyTorch, Hugging Face Transformers, BERT, NLP, Data Preprocessing.

SKILLS

Programming Languages: Python, Java, C.

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, MONAI, OpenCV, LLM, RAG, Agentic AI, Prompt Engineering.

Databases: SQL, MySQL, Relational Databases, MongoDB, Redis, Bigtable, Vector Databases.

Web Technologies: HTML, CSS, JavaScript, React.js, REST APIs, Bootstrap, Streamlit.

Cloud & DevOps: AWS, Linux, Docker, Kubernetes, Jenkins, Git, GitHub, Bitbucket, CI/CD, JIRA, Confluence, Ansible, Maven, Gradle, MLOps.

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01).

AWS Certified Machine Learning Engineer – Associate (MLA-C01).